

Astrodynamics I: Final Exam

June 7, 2017 11:05AM-12:45PM (100 minutes)

Closed Books/Notes

Use the following constants and formulae without proof if necessary:

- Radius of Earth $R_E = 6.378 \times 10^3 \text{ km} = 2.09 \times 10^7 \text{ ft}$
- Gravity Parameter of Earth $\mu_E = 3.986 \times 10^5 \text{ km}^3 / \text{sec}^2 = 1.407 \times 10^{16} \text{ ft}^3 / \text{sec}^2$
- $\vec{e} = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \vec{r} - (\vec{r} \cdot \vec{v}) \vec{v} \right]$, $e = \sqrt{1 + \frac{2Eh^2}{\mu^2}}$

1. (20) A space object enters the atmosphere of the Earth at an altitude of 350,000ft with a velocity of 23,000ft/sec and a flight path angle of -30 degrees.
 - A. (6) What type of orbit does the space object follow?
 - B. (6) What were the velocity vector and flight path angle of the space object when it was at an altitude of 528,000ft?
 - C. (8) Ignoring the effect of the Earth's atmosphere on the motion of the object, determine if the space object impacts the surface of the Earth or not. If it does, what is its impact velocity vector? If not, what is its minimal miss altitude?

2. (30) **Kepler's Equation**

- A. (20) Using the notations in Figure 2-2 and Kepler's Second and Third Laws, derive geometrically Kepler's equation in basic form:

$$\sqrt{\frac{a^3}{\mu}} = \frac{t - T}{E - e \sin E}$$

Hint: You may use the following relations without proof.

$$\begin{cases} BC = B'C\sqrt{1-e^2} \\ PCB = PCB'\sqrt{1-e^2} \end{cases}$$

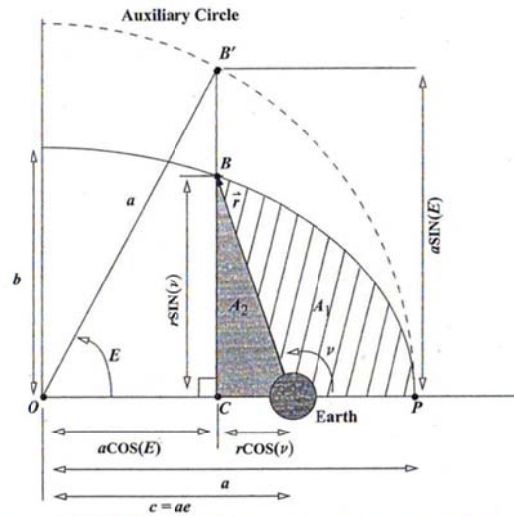


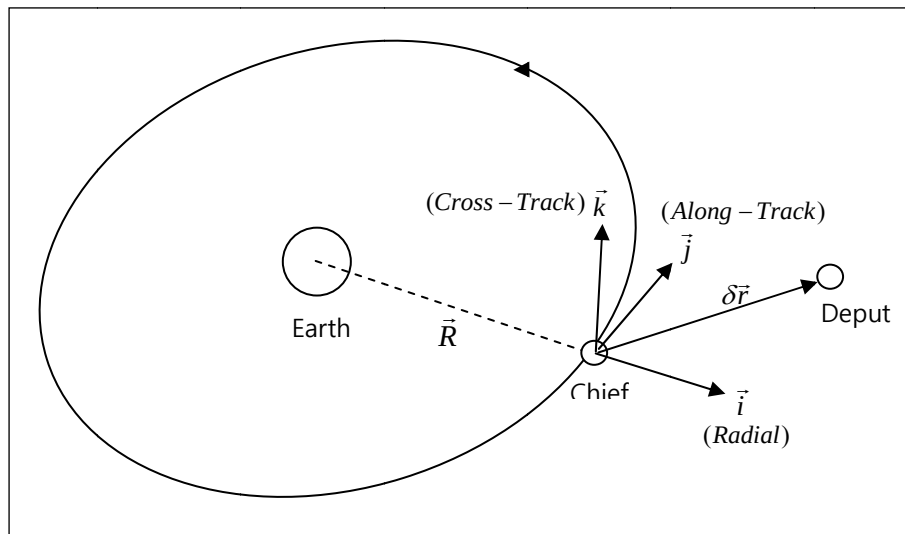
Figure 2-2. Geometry of Kepler's Equation. The eccentric anomaly uses an auxiliary circle as shown. The ultimate goal is to determine the area, A_1 , which allows us to calculate the time.

B. (10) Introduce the notion of mean anomaly $M = E - e \sin E$ and mean motion

$n \equiv \sqrt{\mu / a^3}$ to rewrite Kepler's equation in general form which accounts for (1) multiple revolutions and (2) the initial time **NOT** necessarily located at periapsis.

3. (20) Consider a local-vertical-local-horizontal xyz-frame attached to a circular reference orbit whose constant orbital frequency is given by $\vec{\omega} = \omega \vec{k}$. Let the position of deputy spacecraft with respect to the circular reference orbit (or the chief spacecraft in the figure) be given by

$$\vec{r} = \vec{R} + \delta\vec{r} = (R + x)\vec{i} + y\vec{j} + z\vec{k}$$



A. (5) Determine the velocity and acceleration vector of the deputy spacecraft in terms of

$(\vec{i}, \vec{j}, \vec{k})$ -component.

B. (5) Use the two-body equations of motion

$$\ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r}$$

to derive the equations of motion in terms of $(\vec{i}, \vec{j}, \vec{k})$ -component. Here μ is the gravitational parameter of the Earth and $r = \sqrt{(R+x)^2 + y^2 + z^2}$ is the distance of the deputy spacecraft from the Earth center.

C. (10) Scale (Non-dimensionalize or Normalize) the equations of motion in (b) by using the canonical units, i.e., by using the reference length R and the reference time $1/\omega$.

4. (15) Coordinate Transformations

A. (3) Write down the rotation (coordinate transformation) matrix $R_x(\alpha)$, $R_y(\alpha)$ and $R_z(\alpha)$ about x, y and z-axis, respectively, for an arbitrary angle α .

B. (12) Determine the following coordinate transformation matrices in terms of $R_x(\cdot)$, $R_y(\cdot)$ and $R_z(\cdot)$. It is NOT necessary to expand the products of matrices.

- i. (4) From Geocentric Equatorial Coordinate System (IJK) to Topocentric Horizontal Coordinate System (SEZ)
- ii. (4) From Perifocal Coordinate System (PQW) to Geocentric Equatorial Coordinate System (IJK)
- iii. (4) From Geocentric Equatorial Coordinate System (IJK) to Satellite Coordinate System (NTW)

5. (15) State the definition of the following terminologies succinctly and precisely:

- A. (5) Geocentric Latitude, Geodetic Latitude & Astronomical Latitude
- B. (5) Equation of Time & Equation of Equinoxes
- C. (5) Julian Date